

Day 4 – Client-Server & Peer-to-Peer Architecture

Aim

To understand Client-Server Architecture and build a simple Client-Server network using Cisco Packet Tracer.

Objective

- Understand the concept of Client-Server Architecture.
 - Learn how clients communicate with servers.
 - Configure IP addresses on network devices.
 - Verify connectivity using the ping command.
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Network Topology

PC0 -----

Switch0 ----- Server0

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PC1 -----/

Screenshot 1: Complete Network Topology

[Paste Screenshot Here]

Devices Used

1. PC0
 2. PC1
 3. Switch0 (2960)
 4. Server0
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IP Address Configuration

Device	IP Address	Subnet Mask
PC0	192.168.1.2	255.255.255.0

Device	IP Address	Subnet Mask
PC1	192.168.1.3	255.255.255.0
Server0	192.168.1.10	255.255.255.0

Procedure

Step 1: Add Devices

- Open Cisco Packet Tracer.
- Place two PCs (PC0 and PC1) in the workspace.
- Place one Switch (2960 Switch).
- Place one Server (Server0).

Step 2: Connect Devices

Using Copper Straight-Through cables:

- PC0 FastEthernet0 → Switch FastEthernet0/1
- PC1 FastEthernet0 → Switch FastEthernet0/2
- Server0 FastEthernet0 → Switch FastEthernet0/3

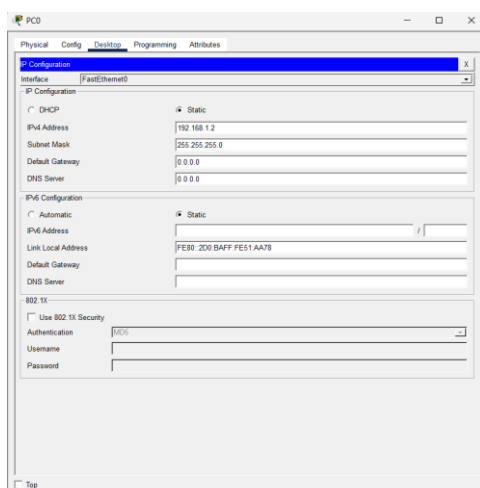
Step 3: Configure IP Addresses

PC0 Configuration

IP Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Screenshot 2: PC0 IP Configuration

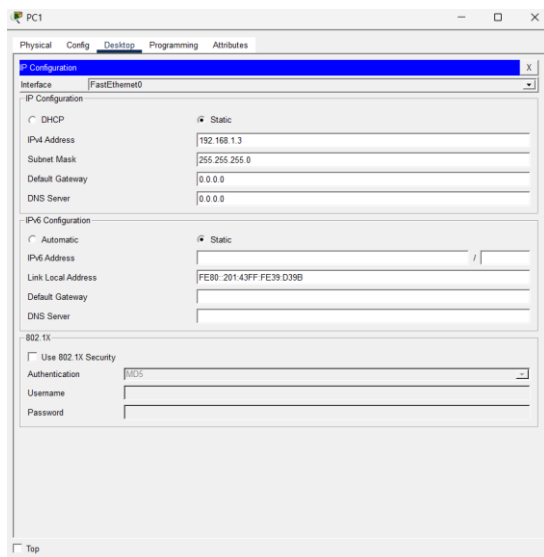


PC1 Configuration

IP Address: 192.168.1.3

Subnet Mask: 255.255.255.0

Screenshot 3: PC1 IP Configuration

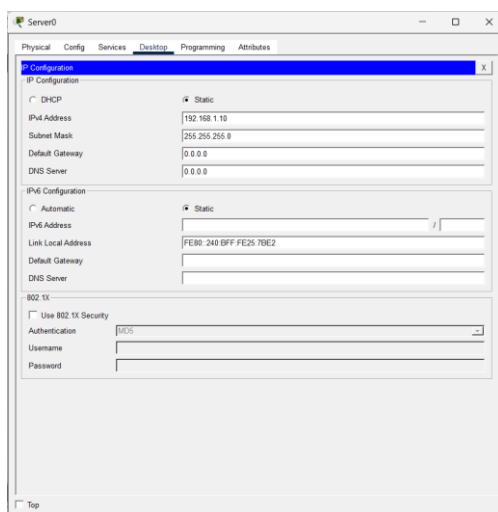


Server0 Configuration

IP Address: 192.168.1.10

Subnet Mask: 255.255.255.0

Screenshot 4: Server0 IP Configuration



Step 4: Verify Connectivity

Open Command Prompt on PC0 and execute:

Command:

```
ping 192.168.1.10
```

Command Used

```
ping 192.168.1.10
```

Expected Output

Reply from 192.168.1.10

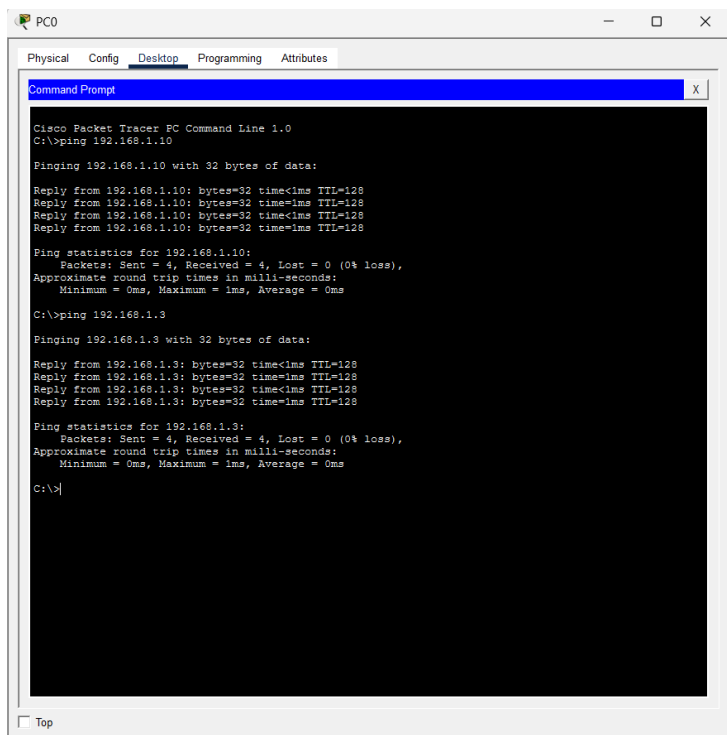
Reply from 192.168.1.10

Reply from 192.168.1.10

Reply from 192.168.1.10

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

Screenshot 5: Successful Ping Result



Observation

- All devices were successfully connected through the switch.
 - The server responded to ping requests from PC0.
 - Communication within the LAN was successful.
 - The switch forwarded packets to the correct destination.
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Result

The Client-Server topology was successfully created and tested in Cisco Packet Tracer. Connectivity between the client (PC0) and the server (Server0) was verified using the ping command.

Conclusion

This experiment demonstrated how Client-Server Architecture works in a network environment. The client initiated communication by sending a request, and the server responded successfully. The switch facilitated communication between devices within the Local Area Network (LAN).

Screenshot 6: Final Working Topology (Optional)

